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An Integrated Analysis Approach for Identifying Water Influx Mechanism in Gas Well: TMG Case Study

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ABSTRACT - Produced water is a primary operational challenge in gas fields, as it can restrict gas flow and severely reduce well productivity. Diagnosing and identifying water influx sources in directional gas wells is critical to mitigating this phenomenon and optimizing well performance. This study details a quantitative workflow evaluating water-gas production dynamics to identify key drivers of water influx, water coning and channeling. Furthermore, it determines whether the reservoir aquifer boundary has shifted. This study uses algorithms derived from classical model water-to-gas ratio diagnostic, reservoir and geological evaluations based on the transient analysis radius of investigation and predictive models to forecast cumulative gas production at breakthrough. By analyzing data from 4 wells, this study demonstrates that water-gas ratio trends and transient pressure behaviors can accurately classify fluid boundary geometries. Driven by localized reservoir heterogeneities and declining pressures, these dynamics accelerate water encroachment. This process culminates in water breakthrough, as confirmed by diagnostic log-log signatures of water channeling and water coning late channeling. The established method provides robust diagnostics for identifying produced water sources and improving predictive accuracy, directly guiding effective well management and field development strategies.

Keywords: gas rate, water rate, drive mechanism, water gas-ratio, water-bearing zone, pressure derivative, heterogeneity, water breakthrough, water influx, water channeling, water coning.

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1. Introduction

One of the major challenges in oil and gas operations is the entry of produced water into the wellbore from hydrocarbon-bearing reservoirs. This encroaching water may be associated with the vertical migration of the water contact toward perforation, a phenomenon known as water coning, or with the rapid intrusion of water from an aquifer that bypasses the hydrocarbon zone, referred to as water channeling. Both phenomena are common and can significantly detract from overall well productivity.

This condition was validated at a research site within the West Java Basin. The TMG field is an active gas asset featuring four production wells with an average daily gas rate (q_g) of 12 MMscfd. This field's reservoir comprises middle shoreface bioclastic carbonate rocks of the Upper Cibulakan Formation (Reksalagora 1996). High localized drawdown gradients established during production have accelerated water production across this sector.

The production history of the TMG field from 2016 to 2025 exhibits a steady increase in dry gas production, followed by a subsequent rise in produced water volume. This behavior stems from declining reservoir pressure, which suppresses gas production while water volumes continue to escalate (Figure 1). These trends support the hypothesis that water influx originates from an active bottom-water or edge-water drive, triggering localized water coning or channeling. Consequently, the affected gas wells lose the critical velocity required to lift liquids, causing fluids to accumulate at the bottom of the borehole and induce liquid loading.

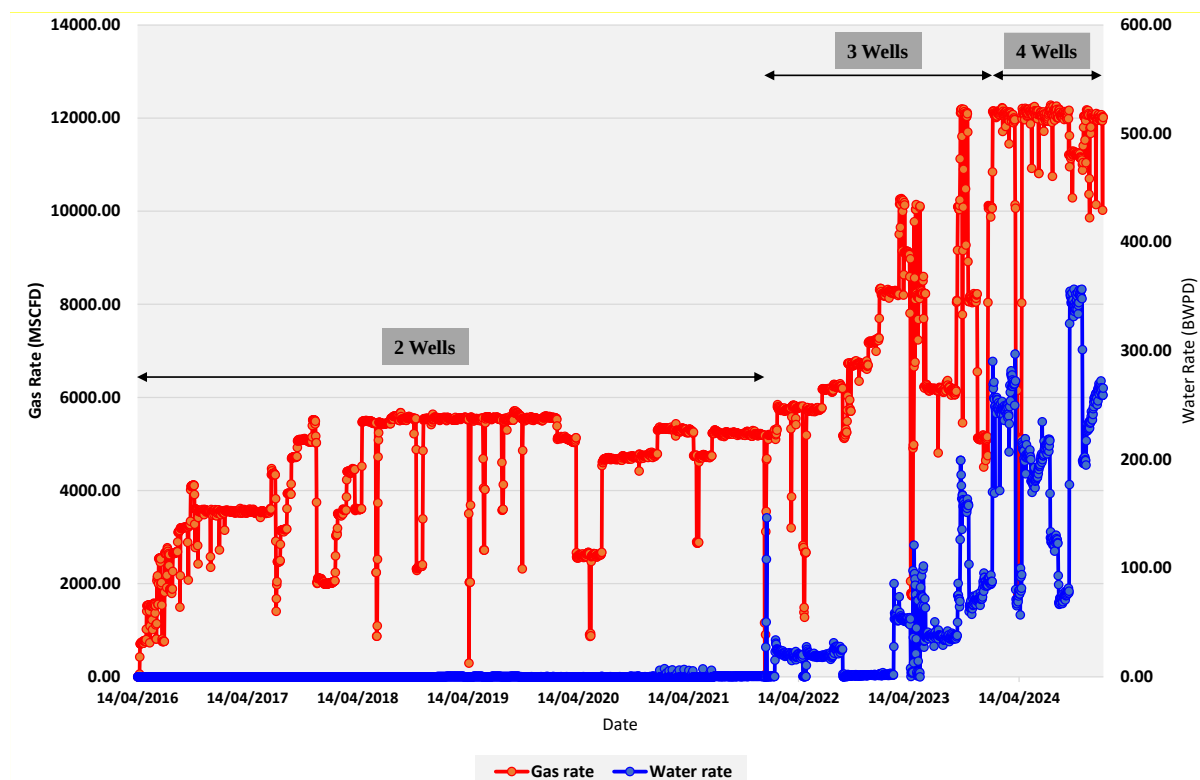


Figure 1.
Profile production of TMG field

Given this context, this research proposes an efficient solution to address the challenges associated with produced water influx in gas wells. This study applies a variety of empirical approaches to determine water production mechanisms, explicitly characterizing active water channeling and coning behaviors in wells experiencing water breakthrough.

2. Methodology

The study methodology is designed as an integrated workflow to identify water production mechanism, ultimately improving well production performance and reservoir development planning. This systematic approach is categorized into four primary stages:

- Data acquisition
- Diagnostic reservoir analysis
- Diagnostic water production from wells
- Reservoir and water production analysis Integration

Figure 2 outlines the integrated workflow for the TMG field case study, which begins with a data acquisition phase. During this phase, quality control measures were applied to reservoir pressures, temperatures, fluid compositions, production histories, core analyses, and well logs to ensure data integrity. This research is divided into two primary sections: reservoir analysis and produced water evaluation. The reservoir analysis investigates reservoir drive mechanism identification, fluid types, fluid flow dynamics, and the prediction of cumulative gas production at breakthrough using a pressure derivative behavior model. Meanwhile, the produced water evaluation determines water influx mechanisms and sources, utilizing water-gas ratio (WGR) analysis to define water-bearing gas zones and identify coning and channeling signatures.

The results of both analyses are combined during the reservoir and water production integration phase. This phase evaluates the water production signatures displayed on the log-log plots, which are characterized as water coning, channeling, or normal displacement. Furthermore, the source of produced water is determined using daily gas and water production histories, log-log water-gas ratio (WGR) plots, and cumulative gas (G_p) versus cumulative water production (W_p) plots. These diagnostics help identify whether the asset is a pore-water, bottom-water, or edge-water gas reservoir, which is subsequently used to predict cumulative gas production at a breakthrough.

The comprehensive sequence of this study is shown in the following flow diagram (Figure 2).

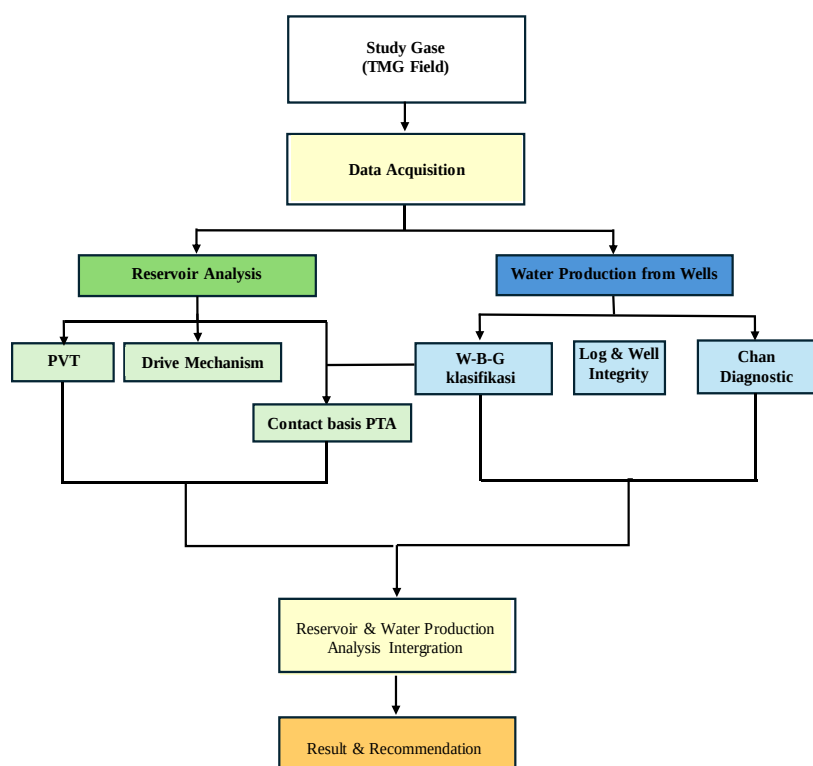


Figure 2.
Flowchart of study

3.1 Data acquisition

The datasets for this research comprise subsurface and production records. The primary inputs include daily gas and water production rates, tubing and wellhead pressures, and cumulative fluid volumes. These are supported by reservoir pressure and temperature measurements, core data (RCAL and SCAL) were incorporated fluid and reservoir properties parameters.

3.2 Diagnostic reservoir analysis

Understanding the significance and types of reservoir drive mechanisms in actively producing gas fields is crucial, as they directly dictate the production behavior of the field under study. Typically, volumetric gas reservoirs are produced primarily through the expansion of the gas contained within the formation. Because natural gas is significantly more compressible than connate water and the underlying aquifer, gas expansion serves as the dominant drive mechanism. However, depending on the characteristics and geometry of the reservoir and aquifer, the primary depletion drive may shift toward a water-drive mechanism with varying drive indices (Fan 2017). A P/Z vs. G_p plot is widely used to identify aquifer water influx. In a purely volumetric reservoir, the relationship between P/Z and cumulative gas production (G_p) is linear; any deviation from this straight line indicates the presence of water intrusion. Once water intrusion is established using the P/Z vs. G_p plot, the next step is to construct a Cole Plot, which provides greater sensitivity for identifying the reservoir's dominant drive mechanism. Figure 3 illustrates the relationship between P/Z and G_p , as well as the corresponding Cole Plot.

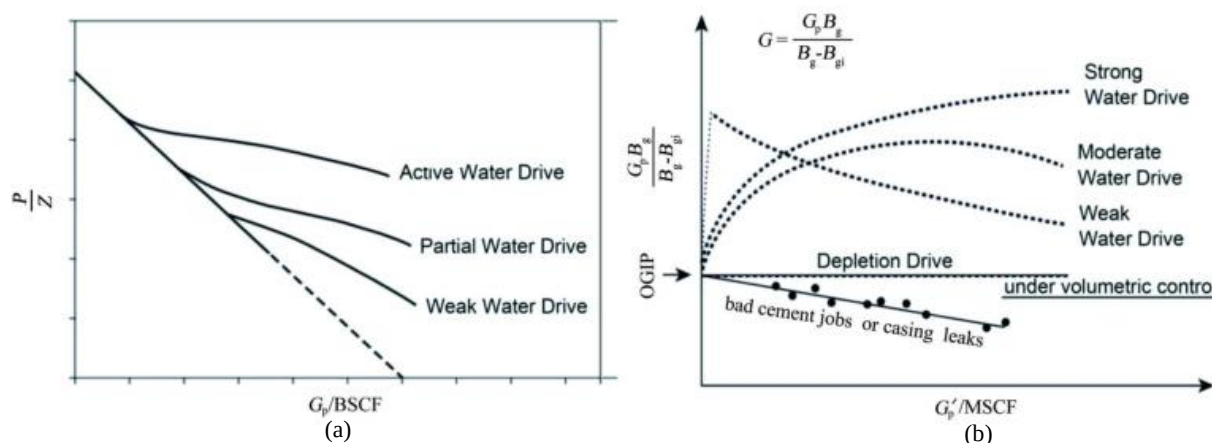


Figure 3

General shapes of P/Z Vs G_p (a) and Cole plot as a function of aquifer strength (a) (Fan 2017)

Another key feature that must be determined in long-term producing gas fields is the classification of the source water-gas zone. This classification can be determined from production behavior. The water-gas ratio approach can categorize produced water sources based on structural geometry into bottom-water configurations with large contact area or edge-water configurations with smaller contact areas (Figure 4); both types are closely related to localized reservoir characteristics (Gao 2023).

Figure 4b illustrates a bottom water gas reservoir, which features a sudden initial increase and a steady rise in the water-gas ratio (WGR) due to a clearly defined underlying aquifer contact boundary. Conversely, if the trend is gradual characterized by a low initial WGR followed by a rapid escalation as gas production declines, and culminating in a stabilized, steady rise at late times the water source is classified as an edge-water gas reservoir (Figure 4a). In this configuration, the water contact boundary is structurally smaller than the reservoir boundary itself.

A major challenge in the gas field is that when a gas reservoir is supported by a strong water-drive mechanism, it drives the migration of the gas-water contact, complicating reservoir development. Yaich (2012) developed a model to detect the presence and position of the fluid contact as a boundary observable through pressure derivative behavior in pressure transient analysis (PTA). Through this approach, the migration of the gas-water contact can

be extrapolated to predict cumulative gas production at the onset of water breakthrough. However, this predictive model must first classify the specific water-bearing gas zone, as the classification dictates the subsequent governing equations. Yaich (2012) proposed two analytical models: the dome model, which accounts for simultaneous lateral water movement and bottom-water encroachment, and the tilted model, which assumes purely lateral water influx. In the tilted model, cumulative production is proportional to the square of the equivalent radius, as illustrated in Figure 5 and Figure 6.

The gas-water contact acts as a constant-pressure boundary within a radial composite reservoir model; thus, if the pressure derivative exhibits a slope deviation, it indicates migration of this fluid interface. This transient behavior is governed by the mobility ratio (M), defined as the ratio of gas mobility to water mobility based on their respective effective permeabilities and fluid viscosities. The mobility ratio can be determined using the Equation 1.

$$M = \frac{(k/\mu)_{gas}}{(k/\mu)_{water}} \quad (1)$$

To quantitatively assess the movement of the GWC, (Yaich 2012) used the equivalent radius, which is a circle with an area equal to that enclosed by the GWC in model. The equation for the equivalent radius is

$$r_e = \sqrt{\frac{A_{GWC}}{\pi}} \quad (2)$$

The area (A) corresponds to a circle of radius (r_i), determined from the radius equation. Quantitative calculation are then applied to extrapolate the cumulative gas prediction at breakthrough using either a conical or cylindrical water-influx model.

$$G = V_{ini} - V_{remain} = \frac{\pi \cdot r_i^2 \cdot h \cdot \phi \cdot (S_{gi} - S_{gr})}{B_g} - \frac{\pi \cdot r_i^2 \cdot h \cdot \phi \cdot (S_{gi} - S_{gr})}{B_g} = A + B \cdot r_i^2 \quad (3)$$

The equation for the conical model can be derived;

$$G = \frac{\pi \cdot r_i^2 \cdot h \cdot \phi \cdot (S_{gi} - S_{gr})}{B_g} - \frac{\pi \cdot r_i^2 \cdot h \cdot \phi \cdot (S_{gi} - S_{gr})}{B_g}$$

Since

$$h_t = r_t \cdot \tan \theta$$

Therefore

$$G = A + B \cdot r_i^3 \quad (4)$$

Identifying water production mechanisms is essential to understand influx sources and optimize reservoir management. To improve well performance and reservoir development planning in the TMG field, this study implements an integrated analytical modeling approach to resolve these operational challenges.

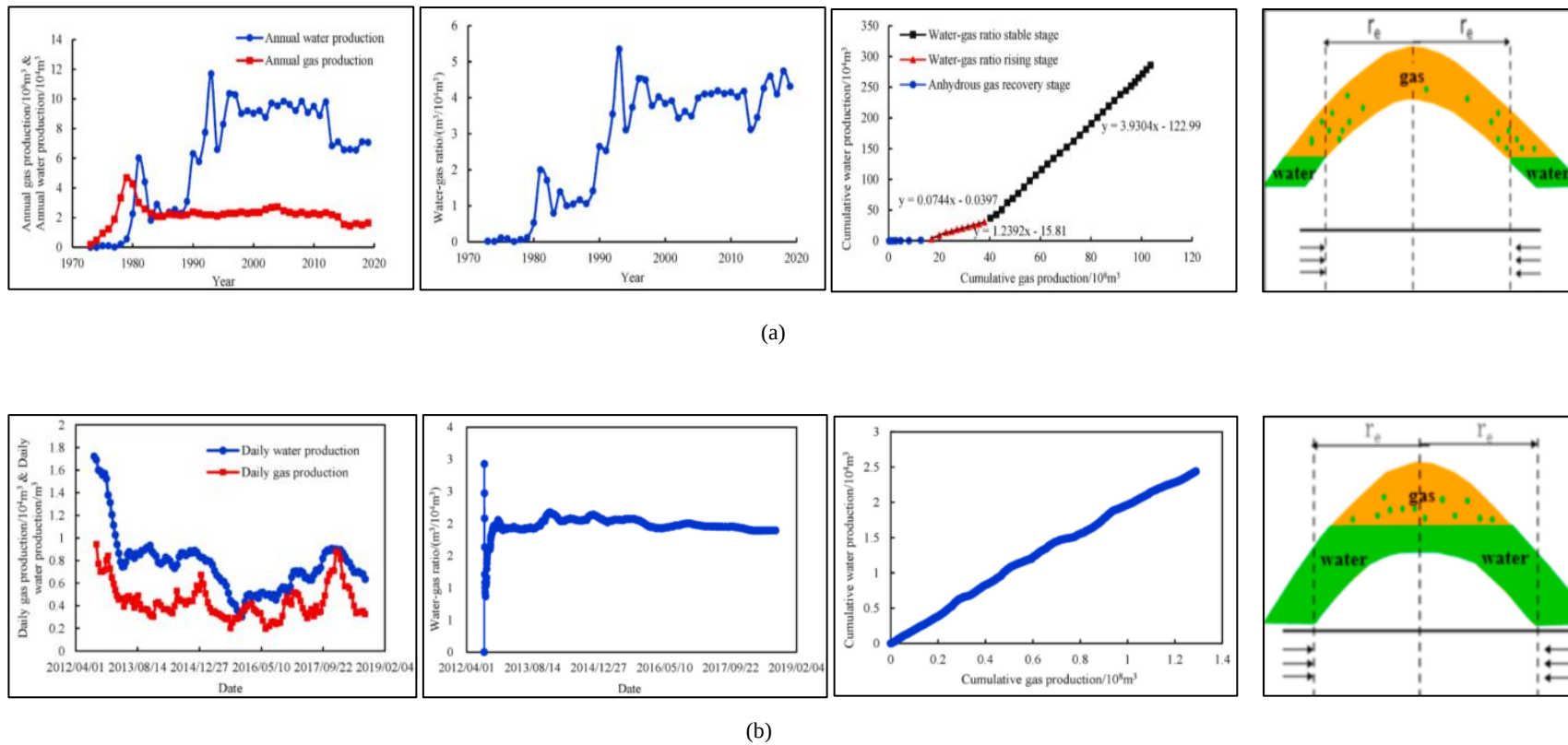


Figure 4.
Water bearing gas zone clasification. Edge water gas reservoir (a) and bottom water gas reservoir (b) (Gao 2023)

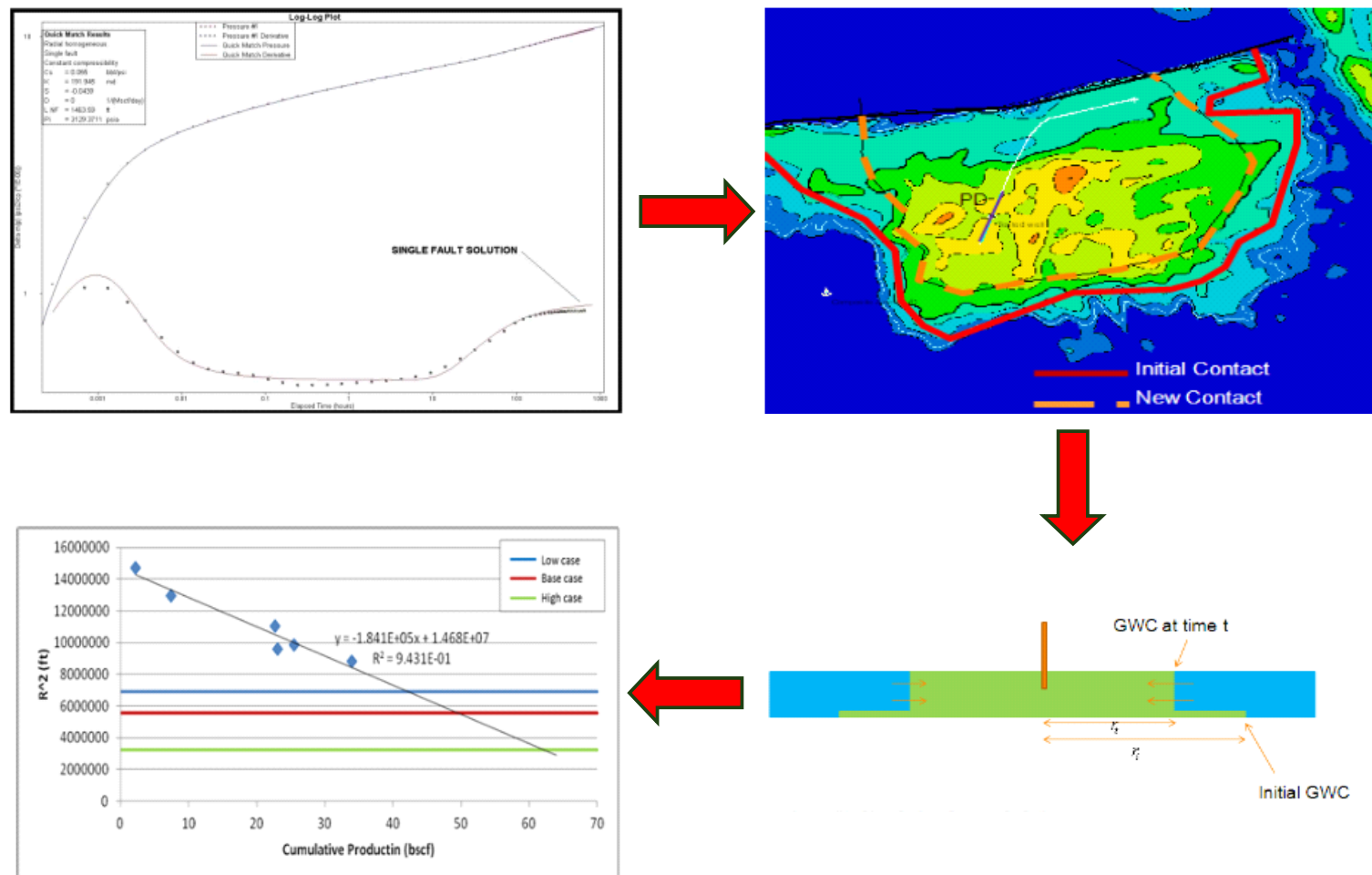


Figure 5.
Tilted model derivative response a (Yaich 2012)

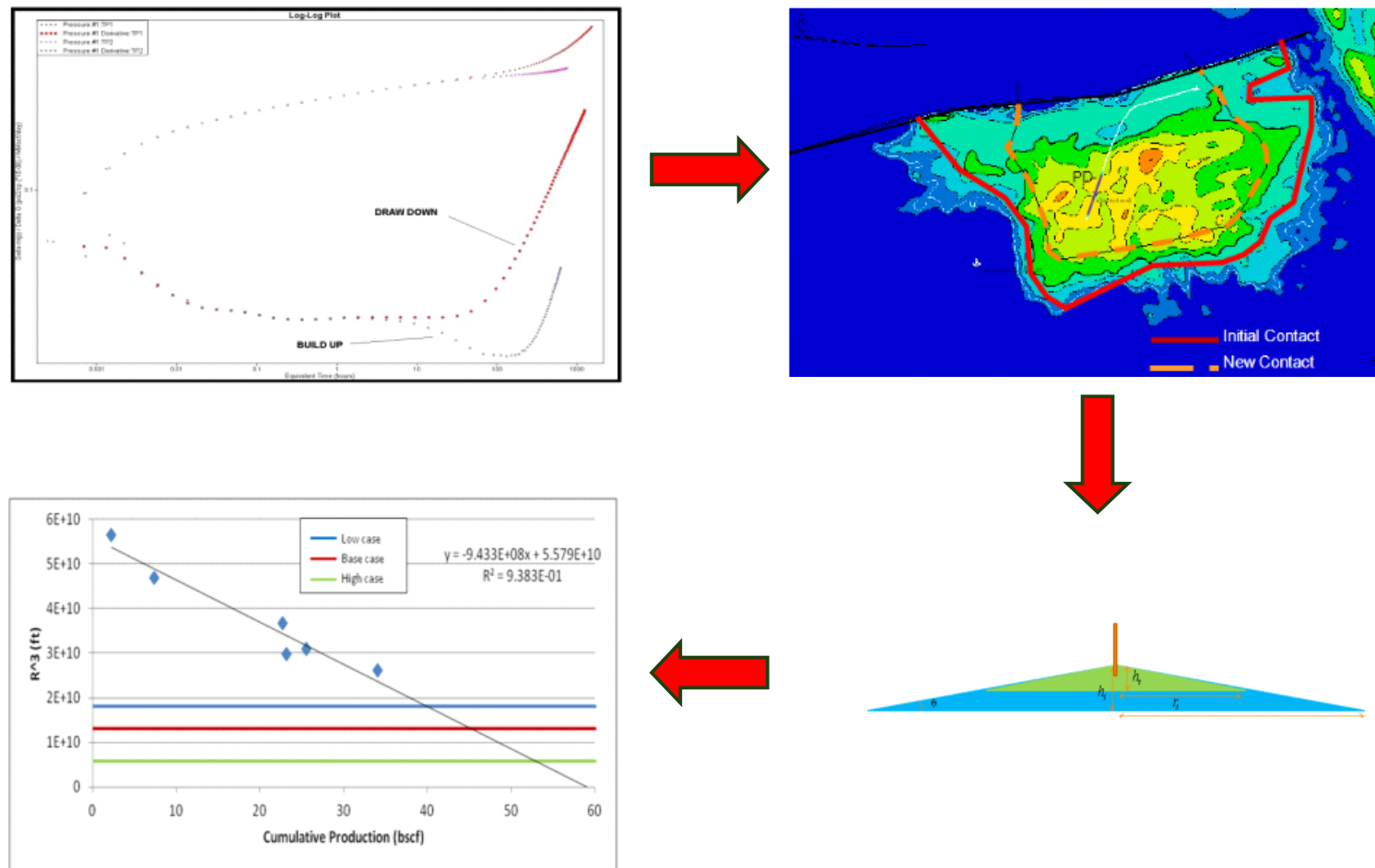


Figure 6
Dome model derivative response (Yaich 2012)

3.3 Diagnostic water production from wells

Al-Ghanim (2010) introduced this approach by expanding Chan's (1995) paradigm. Al-Ghanim (2010) noted that water production sources can be identified using the X-plot technique for production efficiency. Additionally, time-series data of water-to-oil or water-to-gas ratios versus production time, as introduced by Egbe (2005), can be utilized. This is integrated with numerical simulation studies evaluating log-log plots of the water-to-oil and water-to-gas ratios (WOR/WGR) and their time derivatives ($dWOR/dt$ or $dWGR/dt$) to distinguish between coning and channeling mechanisms (Chan 1995). However, in the case study by Al-Ghanim (2010), it was established that Chan's (1995), diagnostic plotting technique provides the most practical and efficient strategy for analyzing rapid water encroachment and selecting precise water control treatments. This approach utilizes the water-gas ratio (WGR) and its derivative (WGR') as primary indicators (Figure 7). The resulting log-log plot categorizes production behavior into normal displacement, water coning, or water channeling signatures based on WGR trends over time, while the WGR' derivative serves to isolate the exact onset of excessive water production.

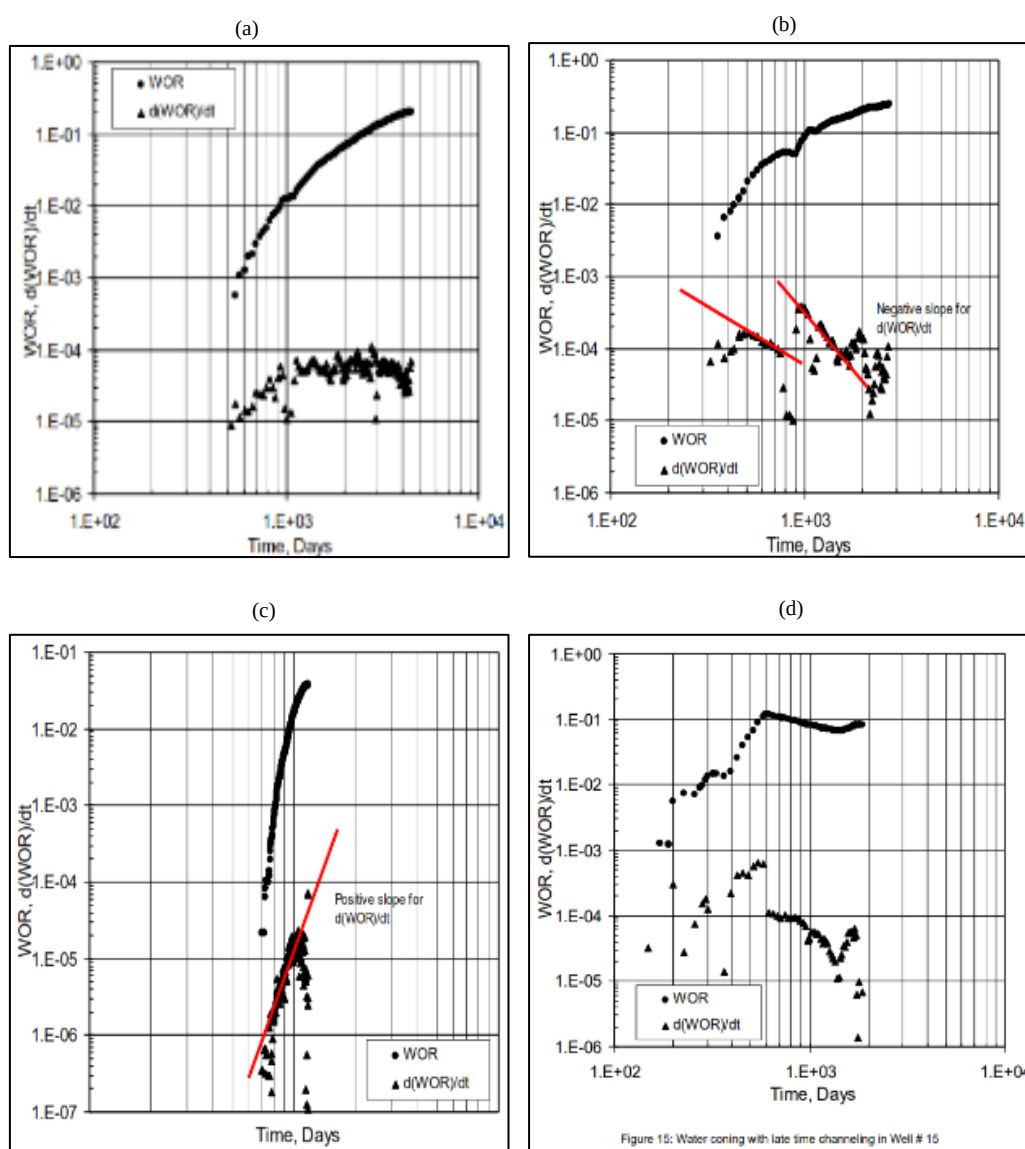


Figure 7

Diagnosing reservoir mechanism based on log-log plot WOR and $dWOR/dt$ Normal displacement (a), water coning (b), water channeling (c), water coning late channeling (d) based on log log plot WGR'/dt (Al-Ghanim 2010).

The mechanism can be diagnosed using the equation for finding the derivative WGR, where

$$\frac{dWGR}{dt} = \frac{WGR_{i+1}(t_i - t_{i-1}) + WGR_{i-1}(t_i - t_{i+1}) + WGR_i(t_{i-1} - t_{i+1})}{(t_{i+1} - t_{i-1})(t_i - t_{i-1})(t_i - t_{i+1})} \quad (5)$$

For normal displacement, the $dWGR/dt$ curve exhibits a flat trend. In contrast, during water coning, the $dWGR/dt$ derivative decreases, showing a distinct negative slope pattern. Finally, a steep positive slope on the derivative curve indicates a water channeling mechanism. During water coning with late channeling, the initial portion of the derivative curve ($dWGR/dt$) exhibits a downward or fluctuating negative slope. The derivative curve subsequently reverses, showing a distinct positive trend at late times (Al-Ghanim 2010).

2.1 Reservoir and water production analysis Integration

The integrated analysis of reservoir behavior and produced water is essential for accurately identifying drive mechanisms, refining original hydrocarbon-in-place estimates, and predicting water breakthrough. This predictive capability enables proactive drawdown management and zone isolation before liquid loading occurs in the wellbore.

3. Result and discussion

A preliminary reservoir analysis was performed to evaluate drive mechanisms using the (P/Z) method. The material balance calculation utilized the gas compressibility factor (z), reservoir pressure (P_i), and cumulative gas production (G_p). The plot of P/Z versus cumulative gas production (Fan 2017) shows a deviates from a straight line (Figure 8), showing the presence of water influx. A Cole plot can be used to determine the type of driving mechanism reservoir, where the cumulative relationship between gas and reserves suggests that the TMG field is moderate-strong water drive.

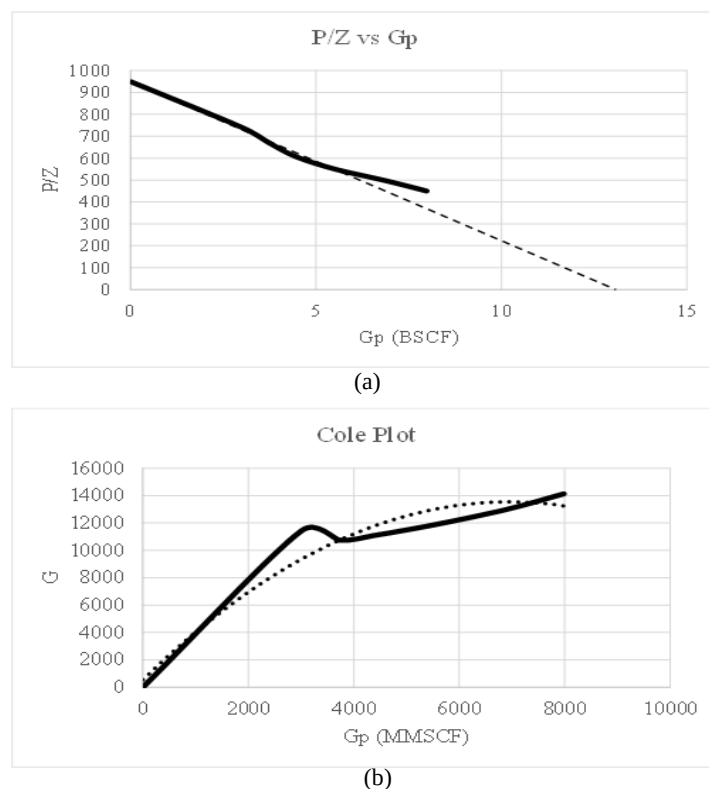


Figure 8
Reservoir drive mechanism in TMG Field.
Presence of water drive based on (a) P/Z vs G_p and strong drive based on (b) Cole Plot

The analysis confirms a moderate-to-strong water drive mechanism, while the steady rise in water production indicates that water influx has already occurred. According to (Gao 2023), water-gas ratio (WGR) behavior can be used to identify water-bearing zones in gas reservoirs. Characterizing these zones plays a critical role in optimizing the development of the TMG field, particularly in improving well placement and perforation design for new wells. Based on the water and gas production behavior in well TMG-03 (Figure 9a), the water-bearing zone was characterized as an edge water-gas reservoir, as evidenced by a gradual, steady increase in the water-gas ratio (Figure 9b). Furthermore, the plot of cumulative gas versus cumulative water production exhibits a distinct two-phase trajectory. Initially, cumulative gas rises rapidly while water production remains negligible a phase known as the “water-free gas recovery stage.” Following water breakthrough, water production escalates while the gas production rate declines, marking the “increasing WGR phase.” Following the increasing WGR phase, the water-gas ratio stabilizes as cumulative water production continues to rise, establishing a linear relationship between cumulative gas and cumulative water during this stable WGR stage (Figure 9c).

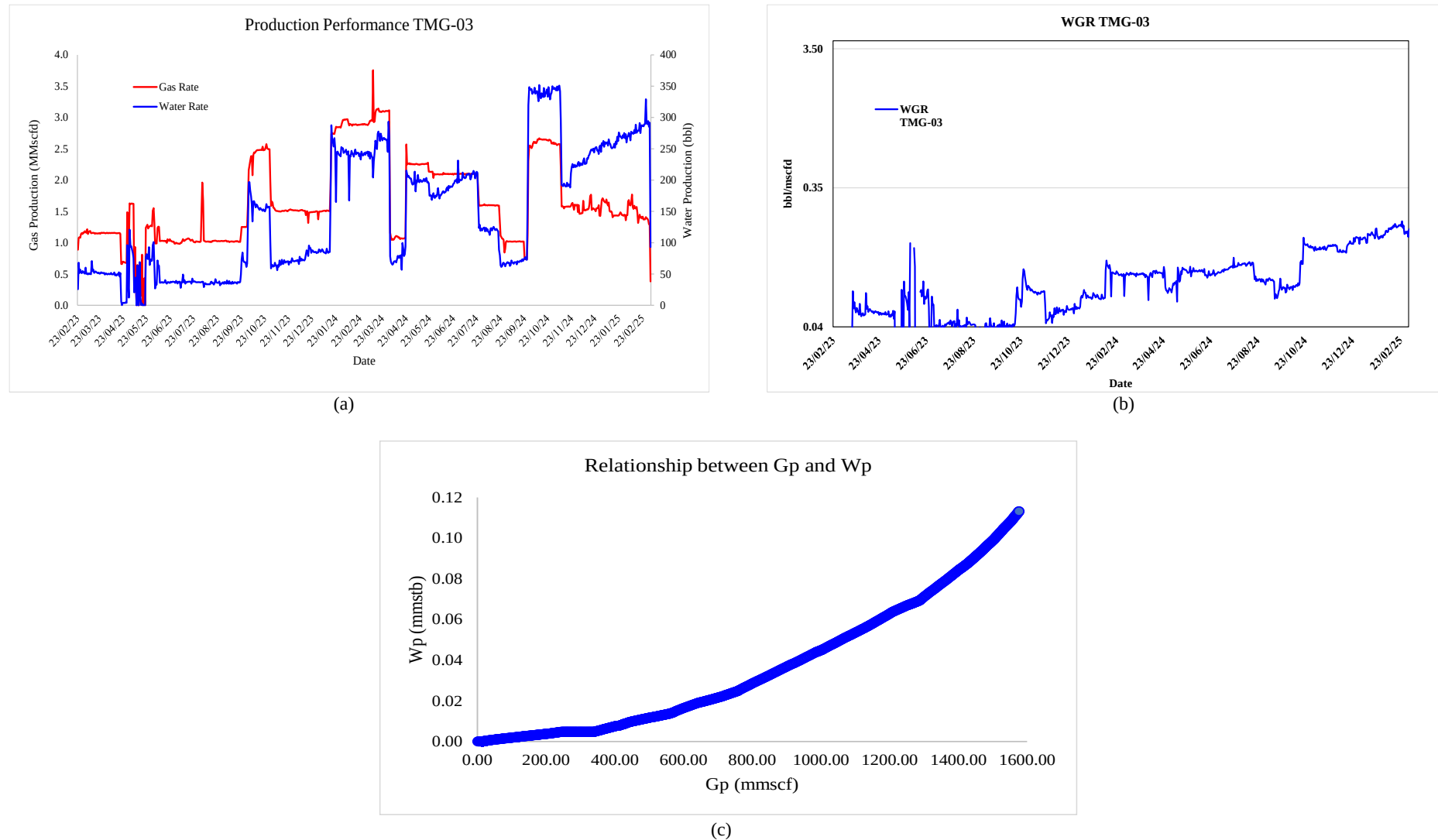
The classification of edge-water gas reservoirs is directly governed by their varied reservoir properties. Based on the porosity-permeability relationship derived from core data, the TMG gas reservoir exhibits significant heterogeneity, divided the formation into multiple distinct rock types. These reservoir characteristics can accelerate the migration of encroaching water into the wellbore. This is further supported by the pressure derivative interpretation, which shows a sharp decline followed by an upward turn in the late-time region, indicating the presence of an active, encroaching aquifer boundary. This investigation was conducted utilizing well test data from well TMG-03 across both the pre-production phase and the subsequent gas and water co-production period.

Initial PBU data (Figure 10a) collected prior to gas production show a sharp downward turn in the late-time pressure derivative. This decline indicates a pressure maintenance boundary, likely marking the initial gas-water contact (GWC). Based on the onset of these boundary effects, the radius of investigation was calculated as 886 ft. According to recent PBU data (Figure 10b), the pressure derivative declines before turning upward in the late-time region; this trend suggests the presence of an outer boundary, such as a fault or localized reservoir heterogeneity. As shown in Figure 11, well TMG-03 is separated from wells TMG-01, TMG-04 and TMG-02 by a fault. Evaluation of the fault seal using the Shale Gouge Ratio (SGR) Equation from (Yielding 2002) reveals a shale ratio of less than 15%, indicating that the fault is non-sealing and may allow fluid communication.

Core data shows that the relationship between porosity and permeability reflects high reservoir heterogeneity. Figure 12 illustrates how the data distribution forms multiple distinct trends or clusters, representing variations in reservoir rock types. When modeling permeability, rock types with overlapping distributions are consolidated into a single Hydraulic Flow Unit (HFU). Consequently, Figure 13 depicts the cross-sectional and vertical distribution of the reservoir permeability and rock type model.

Reservoir heterogeneity directly influences aquifer dynamics; a high permeability path connected to an aquifer with a strong water-drive mechanism accelerates water encroachment. This behavior is reflected in the pressure derivative curve 10b, indicating that the initial gas-water contact (GWC) boundary has shifted. Under these conditions, the radius of investigation was calculated as 694 ft. Figure 14 shows the projection of this calculated radius into the depth structure map.

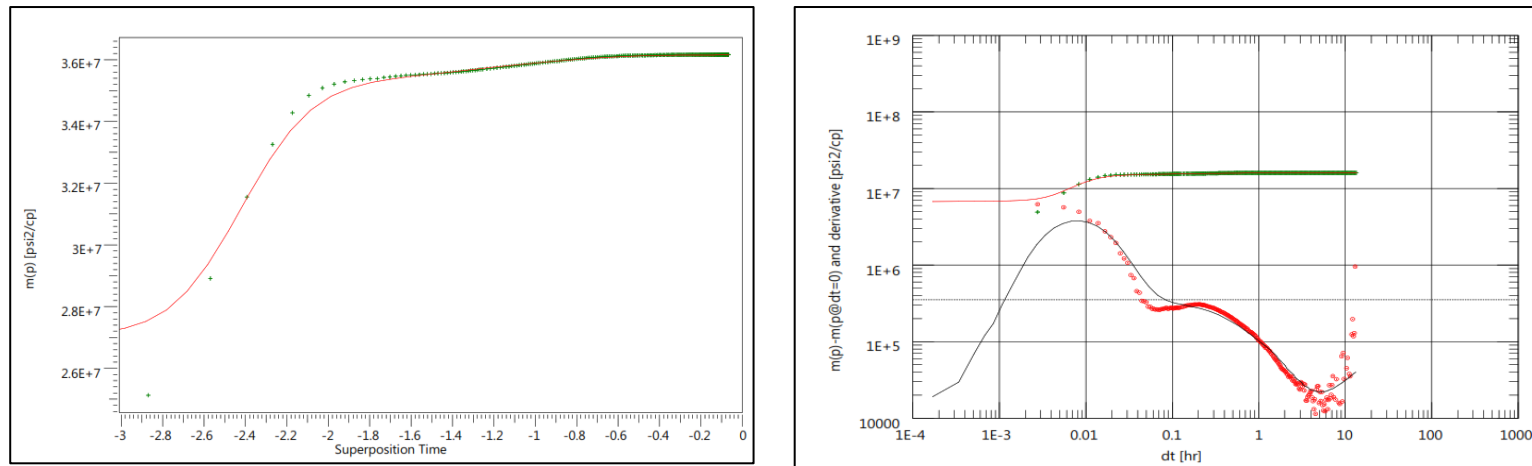
Flow resistance is proportional to the fluid mobility ratio (M). Using Equation 1, this reservoir is calculated to have a mobility ratio (M) of 30, indicating that the water rate more easily than gas. Water breakthrough can be predicted quantitatively using the Yaich (2012) methodology. This predictive model incorporates the classification of the water-bearing zone, the severe water production observed in well TMG-03 and pressure derivative interpretations indicating fluid contact may be shifting. To quantify the movement of the water zone contact boundary, Equation 2 is used to calculate the equivalent radius (r_e), which is then plotted against cumulative gas production (G_p). Assuming a cylindrical geometry (Equation 3), cumulative production scales proportionally with r_e^2 . This model serves to calculate water breakthrough driven by lateral water influx in an edge-water gas reservoir. Table 1 depicts the initial contact boundary at a radius of 886 ft prior to gas production, as well as its progression to a radius of 694 ft during the gas production period.



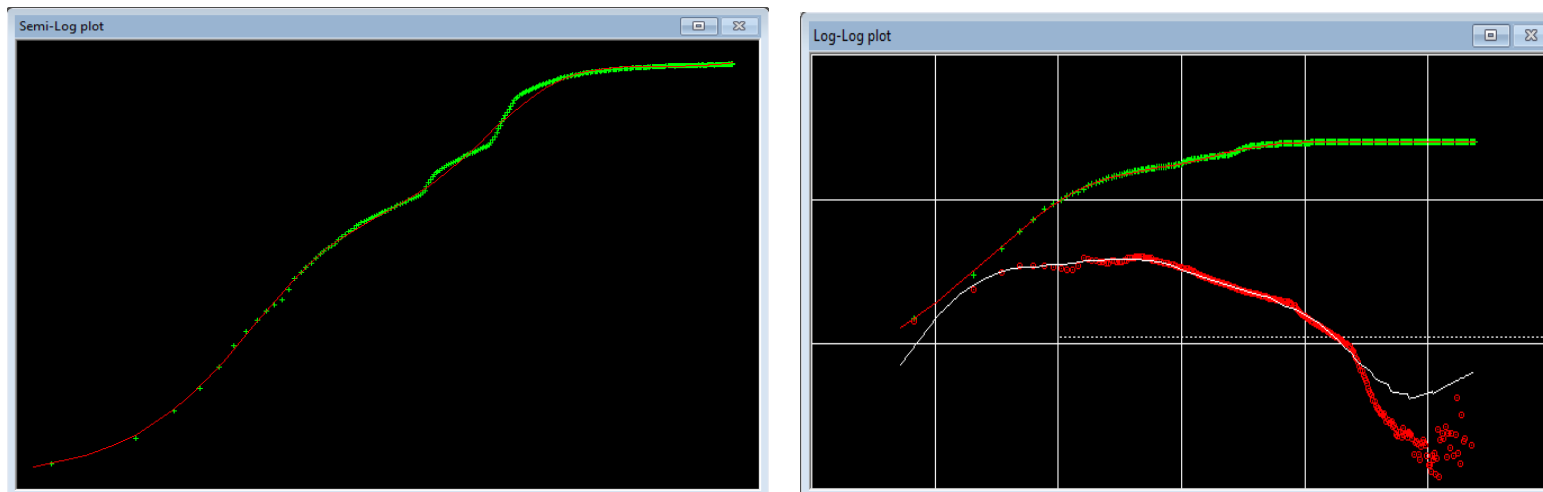
(c)
Figure 9.

Dynamic curves gas and water production in the TMG-03 well.

(a) Curve daily gas and water production, (b) Water gas ratio, (c) relationship cumulative gas (G_p) and cumulative water production (W_p)



(a)



(b)

Figure 10.

The initial pressure derivative curve (a) and the pressure derivative curve from the latest measurement (b) at the TMG-03 well

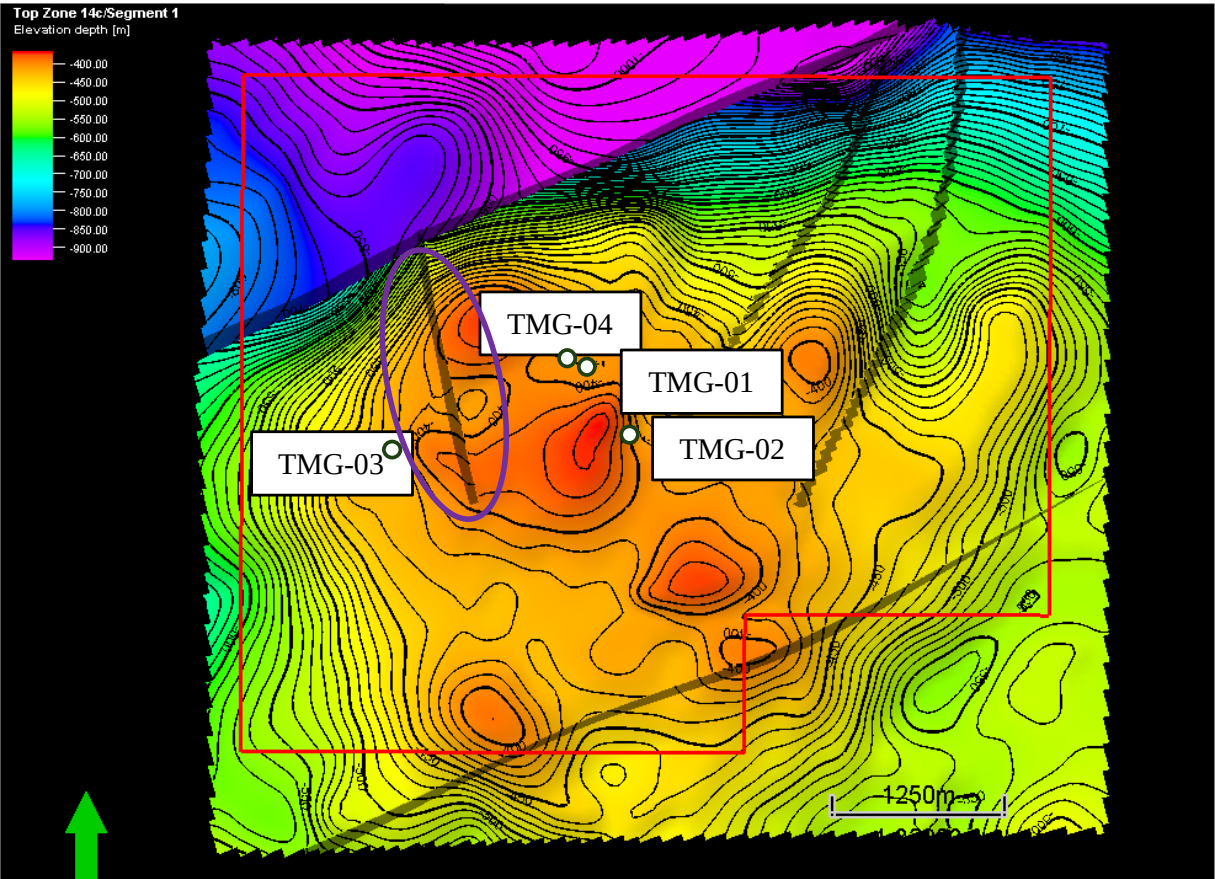


Figure 11.
Fault boundary TMG-03 gas well from other gas wells on depth structure map

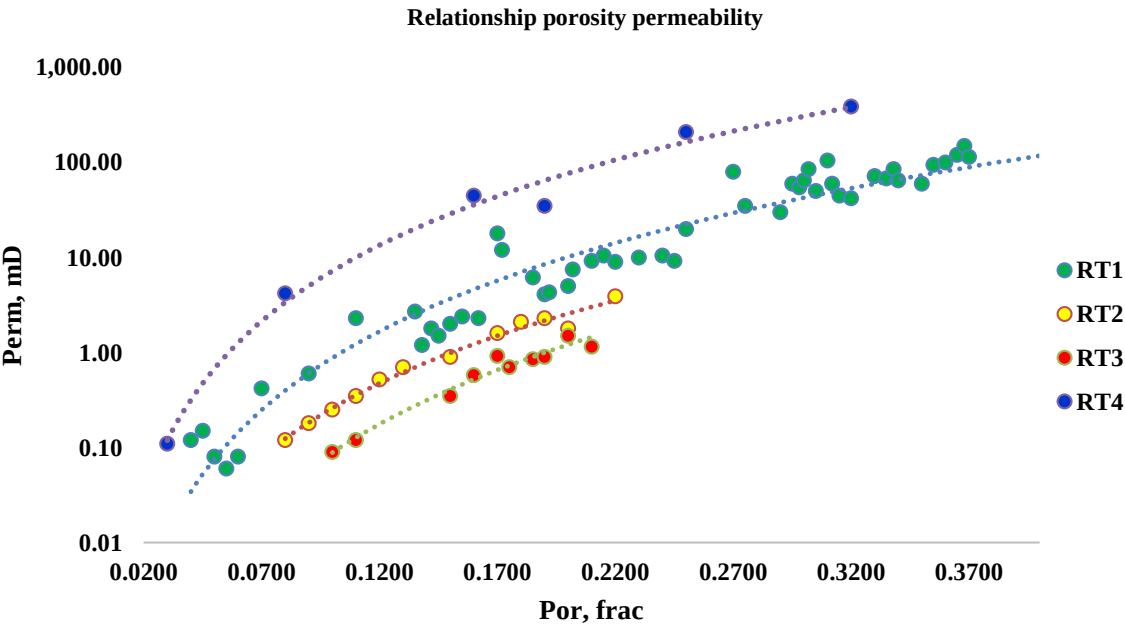
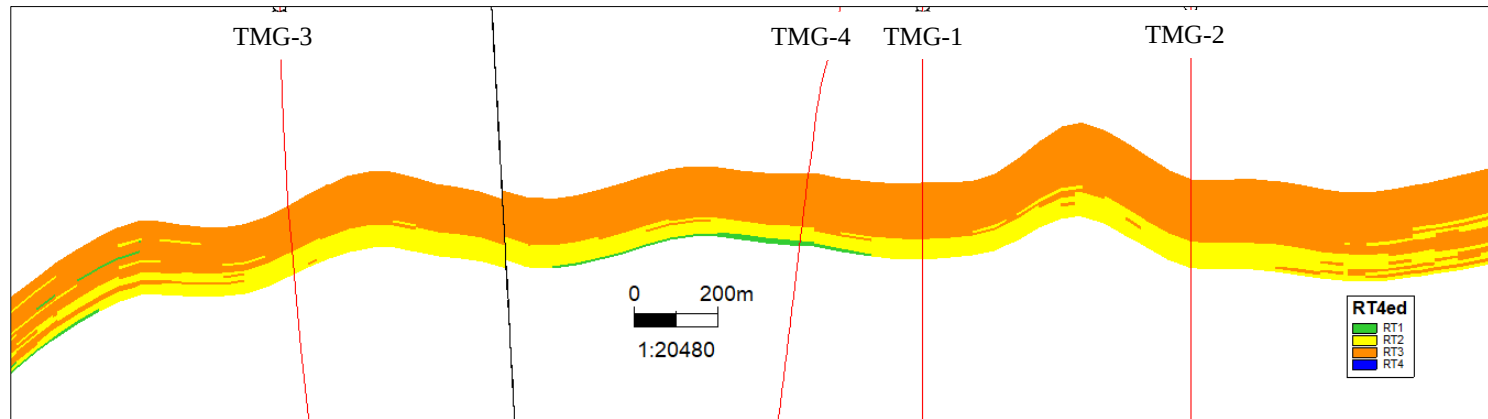
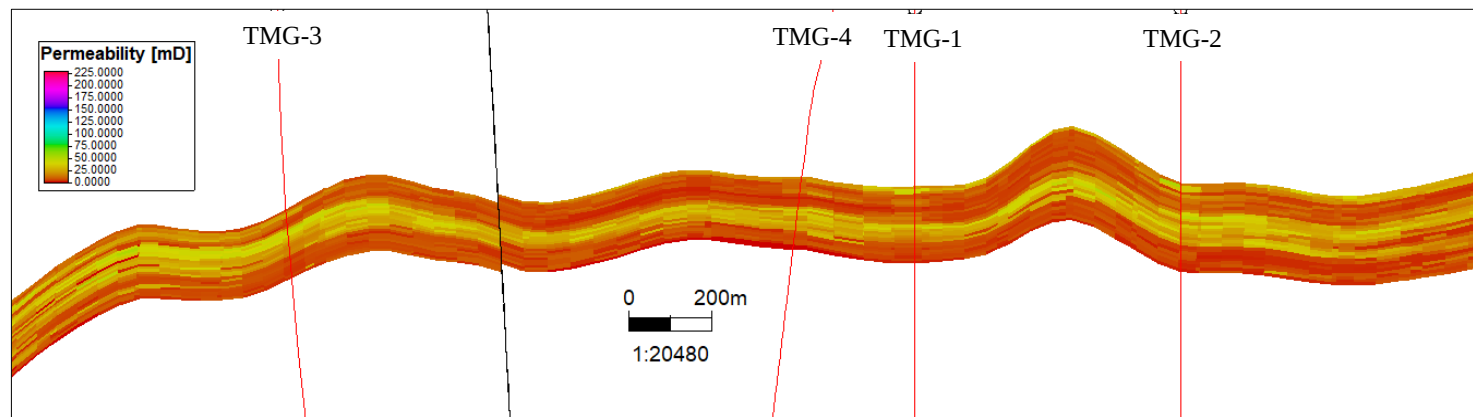


Figure 12.
Crossplot porosity vs. permeability categorized by HFU rock typing method



(a)



(b)

Figure 13.

Cross section of the rock-typing (a) distribution model and the permeability (b) distribution model at the TMG field

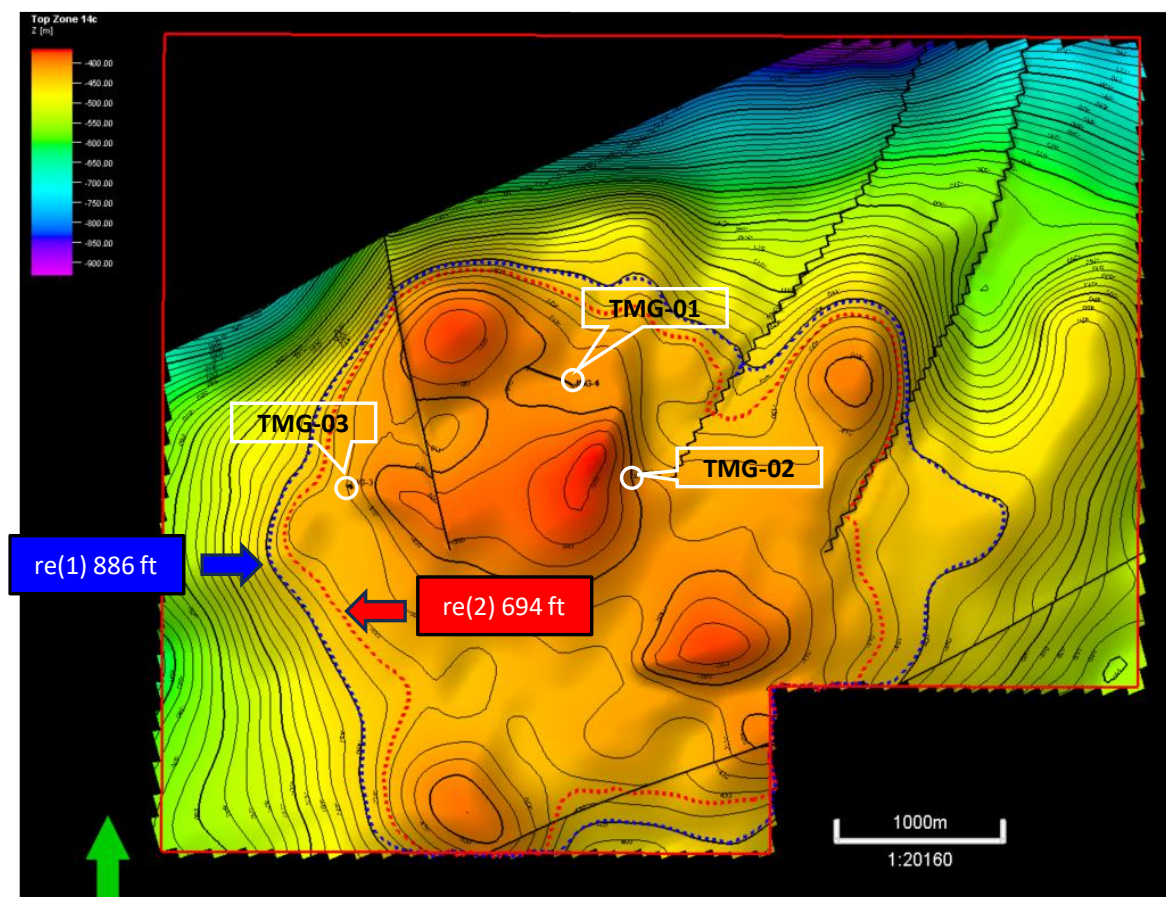


Figure 14.

Initial and new contacts based on radius of investigation calculations from PBU data overlaid with the depth structure map

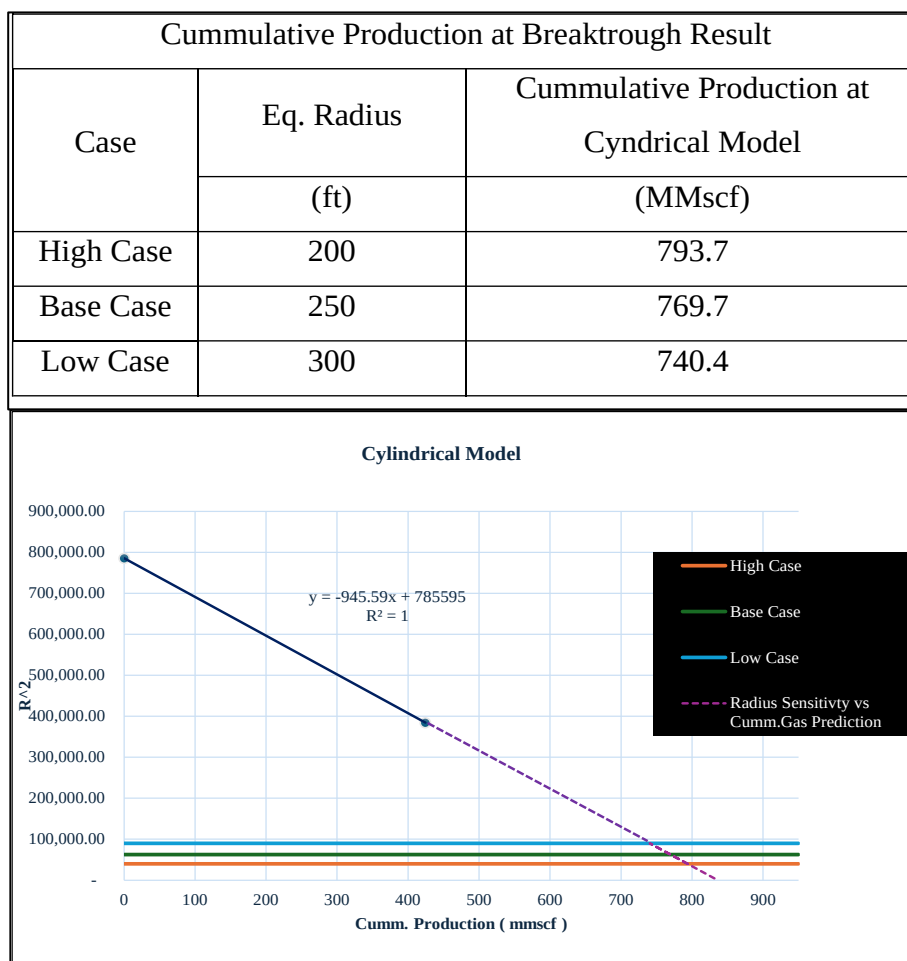
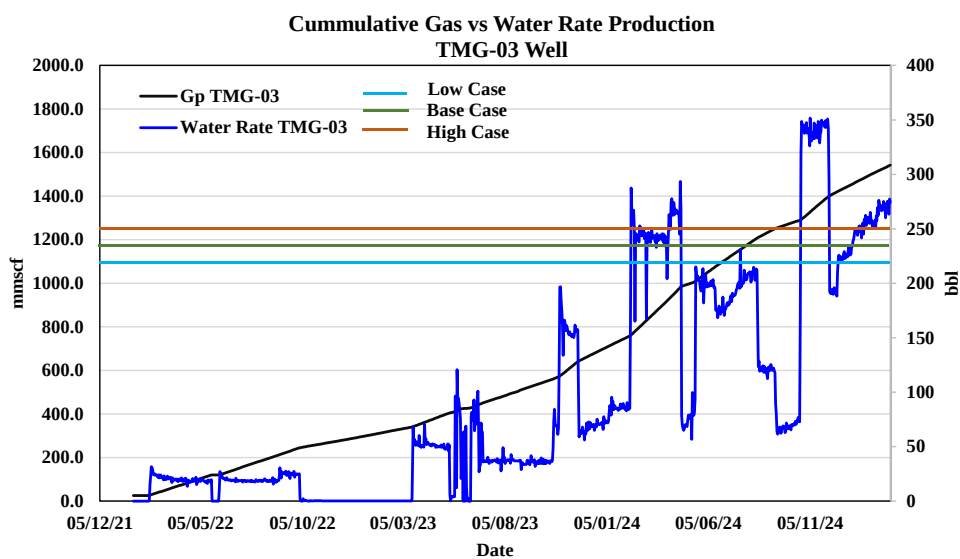
Table 1.
Cummulative gas production at the initial contact versus current contact

GWC distance vs Cum. production TMG-03		
Year Test	Cummulative prodcution (MMscf)	GWC Distance equival ent radius (ft)
Dec-17	0	886.337727
May-23	424.372081	619.928505

According to the Table 1 and Figure 15, water breakthrough is expecated when total cumulative gas production exceeds 11.64 BCF (low case), 11.94 BCF (base case), or 12.18 BCF (high case). As of December 2025, cumulative production from the TMG-03 well stands at 18.37 BCF, surpassing the estimated total cumulative gas production at the time of breakthrough shown in Figure 16. This indicates that the TMG-03 well has already reached the water breakthrough stage.

Water breakthrough is intimately related to the phenomena of water coning and water channeling in the TMG field's production wells. These two phenomena can be identified utilizing Chan's water control diagnostic approach. This approach uses Equation 5, which plots the water-gas ratio (WGR) and its derivative (WGR') against time. The water control analysis includes wells with recorded gas and water production as well as those that have not yet experienced water coning or water channeling.

Figure 15 Cumulative gas prediction at water breakthrough using the cylindrical model

Figure 15.
Prediction of cumulative gas production at the breakthrough time using a cylindrical modelFigure 16.
Cumulative gas production from the TMG-03 well has reached the forecasted value at the time of breakthrough

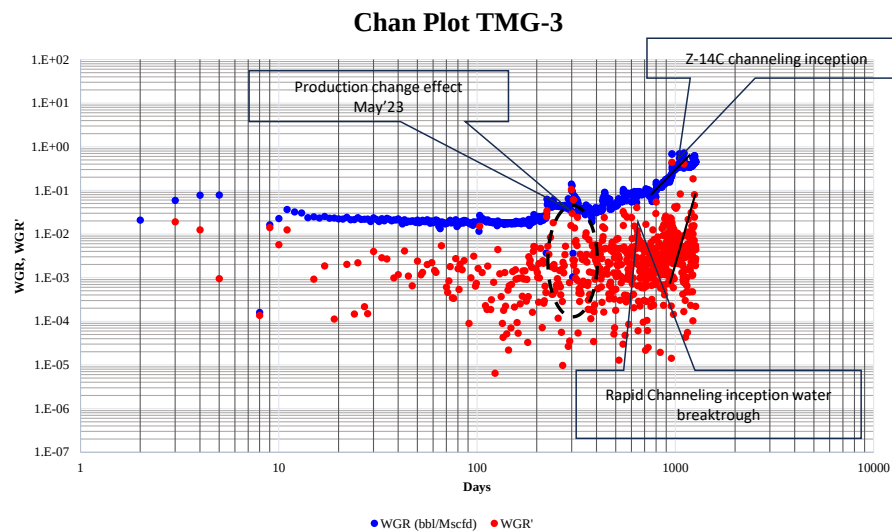


Figure 17
The $dWGR/dt$ plot shows that the TMG-03 well has experienced water channeling.

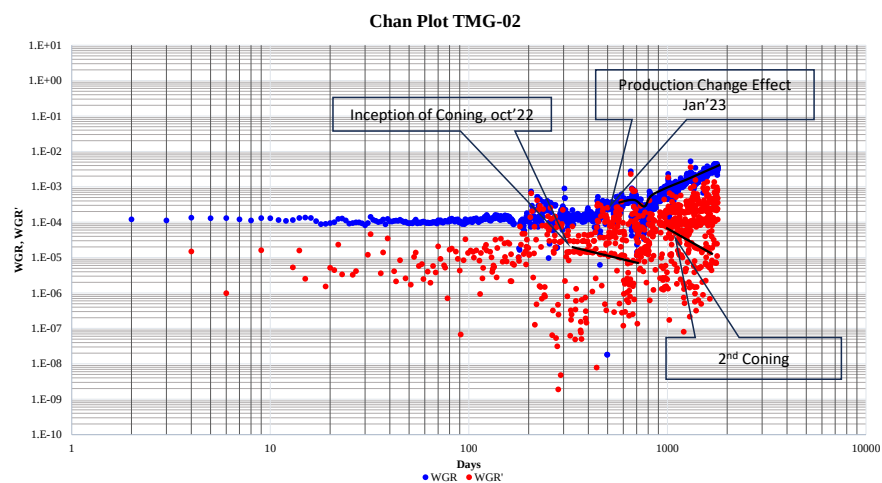


Figure 18
The $dWGR/dt$ plot shows that the TMG-02 well has experienced water coning late channeling.

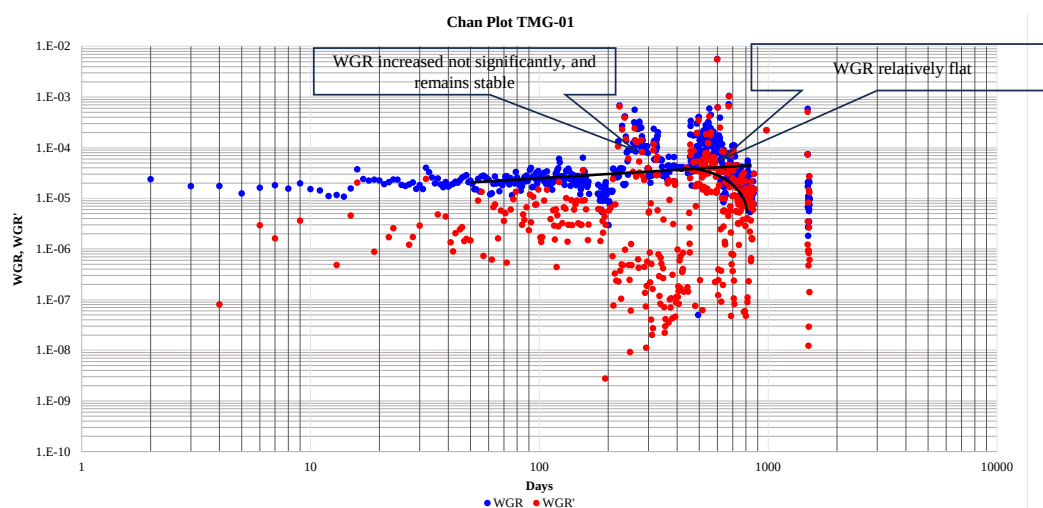


Figure 19
The $dWGR/dt$ plot shows that the TMG-01 well is still exhibiting normal displacement.

The log-log plot of the WGR curve in Figure 17 reveals a high slope, while the WGR derivative has a sharp positive slope, indicating that water channeling and breakthrough have already occurred in the TMG-03 well. Conversely, well TMG-02's WGR derivative originally showed a negative slope but later transitioned to a positive slope by the end of the period, a phenomenon known as water coning late channeling (Figure 18). The primary distinction between the two wells is that TMG-03 has already experienced breakthrough, as evidenced by its very steep positive derivative, whereas TMG-02 exhibits signs of water coning progressing toward channeling. Figure 19 depicts a gradual, steady increase in the WGR over time. The WGR derivative behaves similarly, exhibiting no severe positive or negative slopes, which indicates normal production conditions in well TMG-01.

4. Conclusion

Based on the analysis of the P/Z versus cumulative gas production and the Cole plot, the TMG Field is characterized by a strong water-drive mechanism associated with active water intrusion, as supported by the sustained increase in water production. The observed water influx indicates an edge-water drive mechanism, in which the gas-bearing zone is laterally bounded by an active aquifer. This reservoir behavior is influenced by reservoir heterogeneity and is further supported by the water-gas ratio (WGR) profile, which shows a gradual increase before stabilizing at a relatively high rate. The movement of the gas-water boundary as water production increases is also reflected in the pressure derivative response, characterized by a rapid decline followed by a sharp positive slope in the late-time region, indicating an advancing water front that progressively restricts gas flow. Ultimately, the advancing water front resulted in water influx into the wellbore, as observed in well TMG-03, which exhibited water channeling indicated by a sharp positive slope in the WGR profile, while well TMG-02 showed a transition from early-stage water coning to late-stage water channeling. Overall, these integrated observations consistently demonstrate that active aquifer support and the progressive movement of the water front are the dominant factors controlling the production behavior of the TMG Field.

5. Acknowledgement

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6. Declarations

6.1 Author contributions

Tj. Gede Satryangga performed the data processing, conducted the analysis, developed the methodology, and drafted the manuscript. A. Yasutra and D. Irawan were involved in conceptualization, methodology development, supervision, and reviewed and edited the manuscript. A. Aimar contributed to the water production behavior conceptualization and supervision, and S. Kushidayat contributed to supporting field data record and processed. All authors read and approved the final version of the paper.

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6.3 Competing interest

The authors declare no competing financial interests or personal relationships that could have influenced the work reported in this paper.

6.4 The use of AI or AI-assisted technologies

The author(s) used NotebookLm and Quillbot to assist. After using this tool, the authors review and edit the content as needed and take full responsibility for the publication's content.

7. Glossary of terms and symbols

Terms & symbols	Definition	Unit
q_g	Production gas rate	mscf/day
q_w	Production water rate	bbl/day
t_i	Production time initial	day
Dt	Derivative time	day
K	Permeability	mD
M	Fluid viscosity	cp
R_i	Radius investigation	ft
B_g	Gas formation volume factor	Bbl/mscf
H	Thickness	ft
c_t	Total compressibility	psi ⁻¹
$\emptyset$	Porosity	Frac
A_{GWC}	Contact area	m ²
r_e	Equivalent radius	Ft
r^2	Equivalent radius cylindrical model	Ft
G_p	Cummulative production gas	mmscf
W_p	Cummulative production water	Mmstb
SGR	Shale gauge ratio	%
A, B	Constants used to lump parameters	-
WGR	Water gas ratio	-
WGR'	Water gas ratio	-

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